

Identifying Key Factors for Effective User Requirement Elicitation in Agile and Traditional Software Development: An Empirical Study

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Abstract—Software has become integral to modern society due to rapid technological advancements and expanding business needs. However, the software development industry faces persistent challenges, including project failures, cost overruns, extended timelines, unmet user needs, and quality issues. Ineffective Requirements Engineering (RE) significantly contributes to these failures. Accurately capturing user requirements and adapting to changes throughout the development process remain critical challenges. Stakeholders often struggle to express their needs clearly, and developers may misinterpret or misalign these requirements, exacerbating project risks. This study empirically investigates RE practices in Agile and traditional software development, identifying key factors—client engagement, requirement prioritization, and strategic planning—that enhance effective requirement elicitation and analysis. Based on these findings, we propose a Requirement Definition and Evaluation Model, providing a structured and adaptable framework to manage evolving user needs and enhance project success across various development methodologies.

Keywords— Agile Software Development, Requirements Management, Requirements Gathering, Requirements Assessment, Requirements Prioritization

I. INTRODUCTION

The dynamic nature of business models, increasing competition, evolving stakeholder expectations, and market-driven time constraints pose significant challenges to pre-defined, requirement-based system development [4]. Traditional methodologies often struggle to keep pace with the high rate of requirement changes. Agile software development has emerged as a modern approach in software engineering, providing an alternative to documentation-heavy and rigid development processes. Any time requirements change during the Software Development Life Cycle (SDLC), this methodology works especially well. Challenges with the Analysis and Elicitation of Requirements [15]:

Stakeholder Challenges:

- Users often lack a clear understanding of their requirements and may not have technical expertise.
- They may be reluctant to commit to documented requirements, leading to delays in communication.
- Changes in requirements frequently arise after budgets and timelines have already been established.

- Some stakeholders may either be unwilling or unable to participate in review processes actively.

Developer Challenges:

- Developers may begin coding before fully comprehending the requirements, resulting in defects and rework during testing and verification.
- Technical teams and end-users often use different terminologies, leading to miscommunication.
- Instead of designing a system tailored to client needs, developers may attempt to fit requirements into existing models or frameworks.

II. REVIEW INSIGHTS

This review synthesizes empirical and conceptual findings to identify the key factors and challenges influencing effective requirements elicitation in both paradigms.

A. Traditional vs Agile Requirements Engineering

Traditional requirements engineering (RE) typically follows a linear, documentation-heavy model where requirements are defined upfront and rarely revisited throughout the project lifecycle. This approach offers traceability and control, but lacks the adaptability required in rapidly changing development environments. In contrast, Agile RE emphasizes iterative development, continuous stakeholder feedback, and lightweight artifacts such as user stories and product backlogs [1, 2]. These elements support real-time adjustments to evolving requirements. Batool et al. [3] emphasized that the rigidity of traditional RE limits responsiveness to user needs. Agile, on the other hand, fosters adaptability through iterative cycles and close collaboration, allowing teams to realign requirements based on user feedback. Cao and Ramesh [4] further highlighted Agile's strength in minimizing rework and refining user expectations through continuous engagement.

B. Effective Practices and Persistent Challenges

Agile RE practices such as incremental elicitation, ongoing stakeholder collaboration, and backlog grooming are widely adopted to overcome conventional RE limitations. Inayat et al. [5] identified 17 such practices in a systematic review, including client involvement, iterative updates to requirements, and prioritization techniques. Despite their

effectiveness, persistent challenges remain, especially in globally distributed Agile teams. Limited client availability, fragmented communication, and insufficient documentation frequently hinder alignment. Barata et al. [6] in a survey of Agile professionals, found that while techniques like user stories and backlog refinement are effective, stakeholder communication gaps and lack of formal documentation still impact project outcomes. These challenges indicate the need for structured, yet flexible approaches like the proposed ARCM model to manage continuous changes in distributed Agile environments.

C. Role of User Involvement and Communication

Effective user involvement is a cornerstone of successful Agile requirement practices. Daneva et al. [7] argue that continuous client participation from early stages ensures accurate and evolving requirement articulation. Bano et al. [8] and Tam et al. [9] also noted that low user engagement often leads to misinterpreted requirements, misaligned priorities, and reduced product satisfaction. Agile practices such as frequent reprioritization [4], lightweight documentation, and real-time communication channels help bridge these gaps. Additionally, Agile encourages cross-functional collaboration, reducing ambiguity and improving shared understanding. Bjarnason et al. [10] emphasized that this integrated communication streamlines decision-making and minimizes over-scoping. Techniques like rapid prototyping [11] offer early validation of evolving requirements, further reducing downstream rework and aligning stakeholder expectations.

D. Contemporary Contributions: Models and Frameworks

Recent research efforts have focused on addressing Agile requirements engineering (RE) challenges by proposing targeted models and frameworks. Curcio et al. [13] advocated for just-in-time documentation and the integration of collaborative tools to enhance requirement visibility and traceability, particularly in large-scale Agile settings. Building on this, Koulekar and Ghimire [14] proposed the Agile Requirement Change Management (ARCM) model, specifically designed for global software development environments. Their model emphasizes the critical role of stakeholder involvement, continuous communication, and adaptive processes in managing evolving requirements across distributed teams. The empirical validation of their model underlines its practical value in improving responsiveness and project outcomes in dynamic Agile contexts.

E. Remaining Limitations and Directions for Improvement

Despite advancements, significant limitations persist in Agile requirement practices. While minimal documentation supports agility, it often compromises traceability, especially in regulated industries and geographically distributed teams [4]. Additionally, inconsistent client availability can lead to delayed feedback and misinterpretation of evolving requirements. Marnada et al. [12] highlighted ongoing difficulties in requirement prioritization and communication, particularly within remote and cross-functional teams. These

challenges point to the continued need for more robust stakeholder engagement mechanisms, enhanced collaboration tools, and adaptive frameworks.

III. RESEARCH METHODOLOGY

This study used a survey-based approach to examine user requirement elicitation in Agile and traditional software development. A structured questionnaire, informed by the literature review [5, 3] and Agile principles, was designed to: (1) gather company and RE policy data, (2) assess current RE practices, (3) identify gaps in traditional and Agile approaches, and (4) explore future research directions.

A. Questionnaire Design

The questionnaire comprised four sections:

- Section A- Company Information collected data on company domain, e.g., software development, size, project team size, and Agile method usage, e.g., Scrum. Sample question: "Your organization primarily deals with which domain?".
- Section B1- Current Scenario used a 7-point Likert scale to evaluate RE practices like feasibility studies and requirement clarity. Sample question: "Do you use language simply, consistently, and concisely for describing requirements?".
- Section B2- Agile Perceptions assessed Agile RE practices, such as handling changing requirements and prioritization. Sample question: "Do you think proper user requirements analysis can be done with less documentation?".
- Sections B3 and B4- Gaps and Future Research explored limitations and Agile adoption preferences. Sample question: "Do you think the success rate will rise with the adoption of agile terms?".

The questionnaire was piloted for clarity and used a mix of Likert-scale and open-ended questions for quantitative and qualitative insights.

B. Sample Size Justification

Purposive sampling targeted 75 software companies, with 50 participating (66.6 percent response rate). This sample size, comparable to similar RE studies, supported robust statistical analysis for 12 independent and one dependent parameter (Success). The diverse sample (varying company sizes and domains) enabled the identification of key predictors (Client Engagement, Requirement Prioritization, Project Planning & Strategy) for the proposed Requirement Definition and Evaluation Model.

C. Data Analysis

Data were analyzed using SPSS with ANOVA, Pearson correlation, regression, and descriptive statistics. Cronbach's alpha ($\alpha = 0.753$) confirmed questionnaire reliability. The findings informed a UML-based Requirement Elicitation-Analysis Model. This methodology provides a robust foundation for identifying RE success factors while acknowledging limitations for future research.

IV. RESULTS AND DISCUSSION

This study examined the relationship between 12 independent parameters and one dependent parameter, "Success" (V13), defined by shorter delivery times, higher ROI, meeting customer needs, flexibility, and improved processes. Independent parameters included Workplace Culture (V1), Team Expertise (V2), Individual Attributes (V3), Team Geographical Distribution (V4), Communication & Decision-Making (V5), Client Satisfaction (V6), Client Commitment (V7), Client Engagement (V8), Requirement Prioritization (V9), Team Structure (V10), Quality Assurance & Evaluation (V11), and Project Planning & Strategy (V12).

A. Reliability Test

Cronbach's alpha ($\alpha = 0.753$, Table I) confirmed the questionnaire's reliability, exceeding the 0.6 threshold.

TABLE I STATISTICS ON RELIABILITY

Reliability Coefficient	Cronbach's Alpha Value
The Cronbach's Alpha	.753 (Accepted)

B. Statistical Data Summary

Descriptive statistics (Table II) showed Client Commitment (V7) had the lowest mean (1.27, SD = 0.45357), indicating strong agreement on its importance. Parameters V6, V8, V12, V1, V2, V9, V3, and V5 also fell in the "agree" to "strongly agree" range. Team Structure (V10) and Geographical Distribution (V4) had higher means, suggesting moderate agreement. All mean scores were below 3 on a 7-point Likert scale, indicating general agreement.

TABLE II SUMMARY OF STATISTICAL DATA

Parameter	Sample Size (N)	Min Value	Max Value	Average (Mean)	Standard Deviation
V1	50	1.00	2.40	1.5660	.38468
V2	50	1.00	3.00	1.6700	.62173
V3	50	1.00	3.60	1.9220	.55245

V4	50	1.00	3.50	2.3500	.63386
V5	50	1.00	4.00	1.9300	.76792
V6	50	1.00	3.00	1.4200	.53552
V7	50	1.00	2.00	1.2700	.45357
V8	50	1.00	3.00	1.3600	.56739
V9	50	1.00	2.50	1.6320	.41468
V10	50	1.33	5.33	2.5674	.86517
V11	50	1.00	3.00	2.0240	.40525
V12	50	1.00	3.00	1.3300	.51260
V13	50	1.20	2.80	1.9620	.43884

C. Correlation Examination

Correlation analysis (Table III) identified significant relationships ($p < 0.05$) between Success (V13) and V1, V3, V6, V7, V8, V9, V11, and V12. V2, V4, V5, and V10 showed no significant correlation.

TABLE III RESULT OF CORRELATION ANALYSIS

Parameters	Correlation Coefficient	Significance (2-tailed)	Interpretation (At the 0.05 level, significant or not)
V1	.302	.035	Yes
V2	.175	.231	Not
V3	.317	.025	Yes
V4	-.254	.077	Not
V5	.236	.096	Not
V6	.432	.003	Yes
V7	.408	.002	YES
V8	.436	.003	Yes
V9	.331	.008	Yes
V10	-.013	.975	Not
V11	.295	.035	Yes
V12	.423	.003	Yes

TABLE IV SUMMARY OF REGRESSION MODEL

Respected version	R Value	R ² Value	Adjusted R ²	Standard Error
I	.447 ^a	.181	.164	.39793
II	.587 ^b	.367	.339	.35852
III	.664 ^c	.438	.381	.34354
a. Forecasters for I: V8. b. Forecasters for II: V8, V9 c. Forecasters for III: V8, V9, V12				

TABLE V: SUMMARY OF REGRESSION COEFFICIENTS

Version	Unstandardized Coefficients		Standardized Coefficients	T-Value	Significance	Correlations			Collinearity Statistics	
	B Value	Standard Error	Beta Value			Zero-order	Partial	Part	Tolerance	VIF
1	1.516	.140		10.053	.001					
V8 constant	.328	.100	.427	3.355	.002	.437	.436	.437	1.000	1.000
2	.726	.254		2.720	.009					
V8										
V9	.380	.082	.515	4.245	.001	.437	.526	.497	.973	1.018
constant	.437	.136	.423	3.469	.000	.330	.452	.407	.973	1.026
3	.562	.250		2.168	.035					
V8										
V9	.341	.090	.429	3.768	.000	.427	.486	.421	.918	1.078
V12	.381	.132	.351	3.128	.002	.340	.430	.350	.939	1.075
Constant	.249	.089	.289	2.402	.021	.412	.332	.258	.924	1.081
a. Dependent Parameter: V13										

TABLE VI: ANOVA RESULTS

Version	Source of Variation	SS	Df	MS	F-Statistic	Significance (p-value)
I	Model (SSR)	2.215	1	2.215	9.871	0.003 ^a
	Error (RSS)	8.126	49	0.166	-	-
	SST	10.341	50	-	-	-
II	Model (SSR)	3.945	2	1.972	12.534	0.001 ^b
	Error (RSS)	6.396	48	0.133	-	-
	SST	10.341	50	-	-	-

III	Model (SSR)	5.187	3	1.729	14.012	0.000 ^a
	Error (RSS)	5.154	47	0.110	-	-
	SST	10.341	50	-	-	-

Regression Sum of Squares (SSR): This represents the variability in the dependent variable that is explained by the regression model.

Residual Sum of Squares (RSS) / Sum of Squares Error (SSE): This represents the variability in the dependent variable that is not explained by the regression model.

Total Sum of Squares (SST): Represents the overall variability in the dependent variable.

SS (sum of Squares)

Df (Degree of Freedom)

MS (Mean Square)

Forecasters for I: V8.

Forecasters for II: V8, V9

Forecasters for III: V8, V9, V12

Dependent Parameter: V13

D. Multivariate Regression Analysis

The regression model summary is shown in Table IV. The R values for each model, which display the strength of the different correlations, indicate the level of linear connection between the actual and predicted outcomes of V13 "Success." Higher R values indicate stronger correlations, and these values range from 0 to 1. Regression models (Table IV) showed Model 3, with predictors V8, V9, and V12, had the highest R² (0.438) and adjusted R² (0.381), explaining 43.8% of the variance in Success.

The t-statistics, which establish the relative significance of each parameter in the model, are shown in Table V. T-values below -2 or above +2 are often regarded as indicators of statistical significance. Table V confirmed significant contributions of V8, V9, and V12 (t-values > 2, p < 0.05), with no multicollinearity (VIF < 2).

E. ANOVA Analysis

The regression analysis's ANOVA results, which show the variance distribution among the three models, are shown in Table VI. While the residual row shows the variation that cannot be explained, the regression row shows the percentage of variance that each model can account for. The sum of the regression and residual variances yields the total variance. Model 3's higher explanatory ability is demonstrated by its lowest residual variance and largest regression sum of squares among the three models. Since the corresponding p-value (.000) is significantly below the 0.05 threshold for statistical significance, the F-statistic results validate that the independent parameters V8 (Client Engagement), V9

(Requirement Prioritization), and V12 (Project Planning & Strategy) are statistically significant predictors of success.

F. Requirement Definition and Evaluation Model

The proposed Requirement Definition and Evaluation Model aims to enhance the requirement engineering phase through a structured, methodology-agnostic approach, based on empirical predictors: Client Engagement, Requirement Prioritization, and Project Planning & Strategy. It uses three levels of abstraction for clarity:

- Level A (Fig. 1): Outlines the core tasks of the requirement engineering process.
- Level B and C: Provide detailed breakdowns of key parameters and sub-tasks.

1) B1 Level: Strategic Requirement Definition

This phase refines stakeholder requirements, transitioning from high-level needs to detailed specifications through steps like problem vision and requirement description. The output is a Finalized Requirement Backlog, dividing requirements into manageable parts (Fig. 2).

2) B2 Level: Requirement Analysis and Prioritization

This phase compiles requirements from the Finalized Requirement Backlog, assesses their value, and assigns priorities collaboratively, creating a Prioritized Requirement Backlog. A graphical Requirement Specification is developed for customer feedback. The project manager then selects high-priority requirements

for the next cycle, forming a Sprint Development Backlog, with clients validating the Deployment Plan and backlog (Fig. 3).

3) Level C: Detailed Breakdown

- Requirement Value Assessment: Evaluate each requirement's value based on specific criteria (Fig. 4).
- Requirements Prioritization Framework: Assigns priorities using influencing factors (Fig. 5).

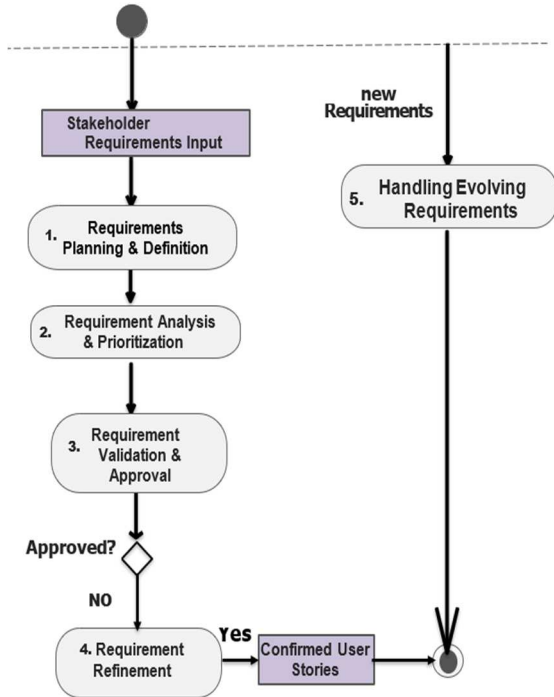


Fig 1. A-Level: Engineering workflow for software requirements

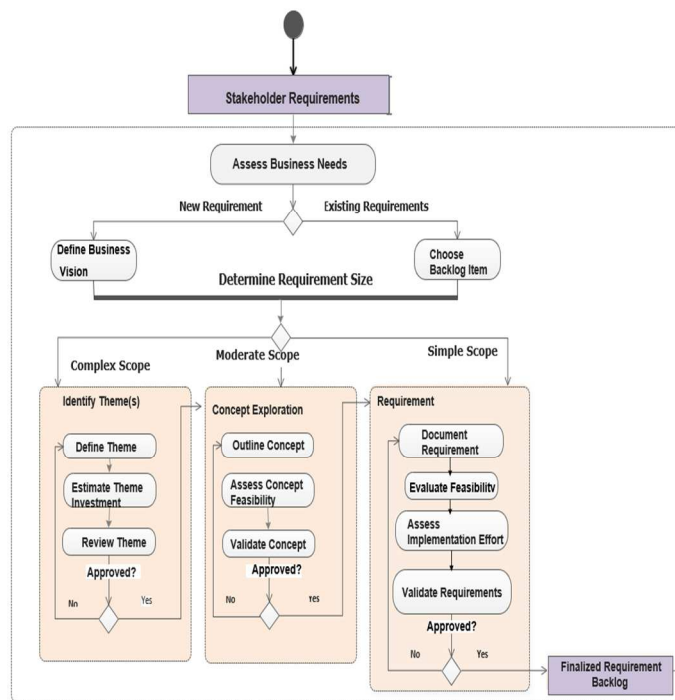


Fig 2. Strategic requirement definition

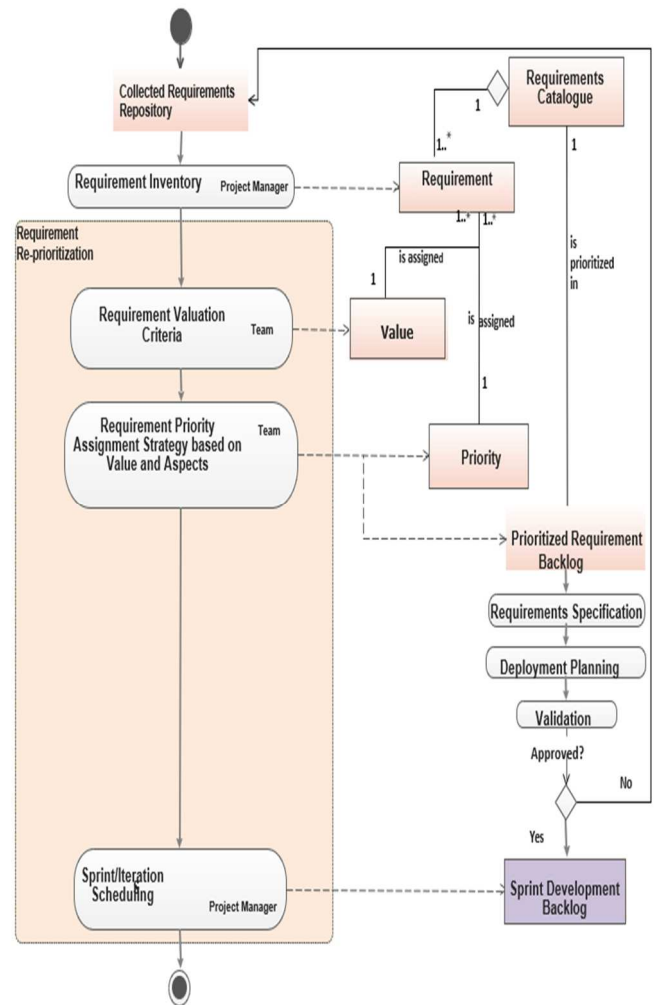


Fig 3. Framework to analyze / Prioritize requirements

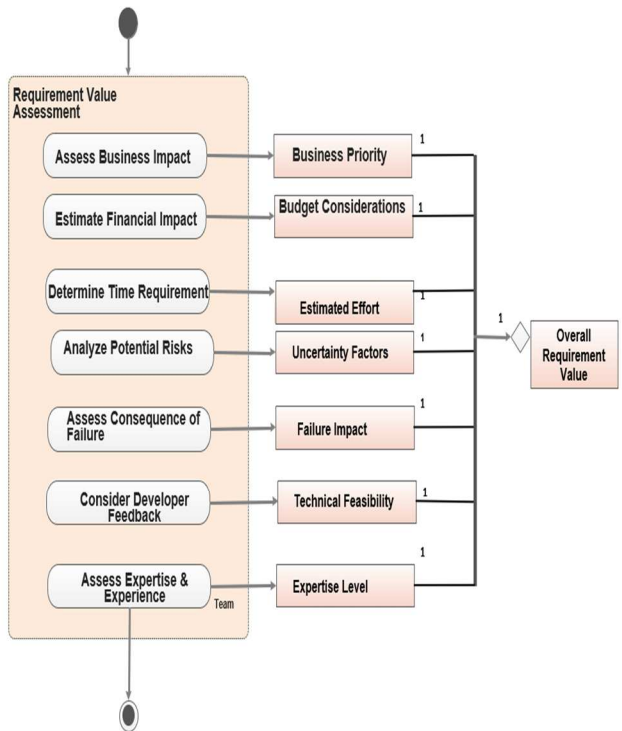


Fig 4. Framework for value estimation

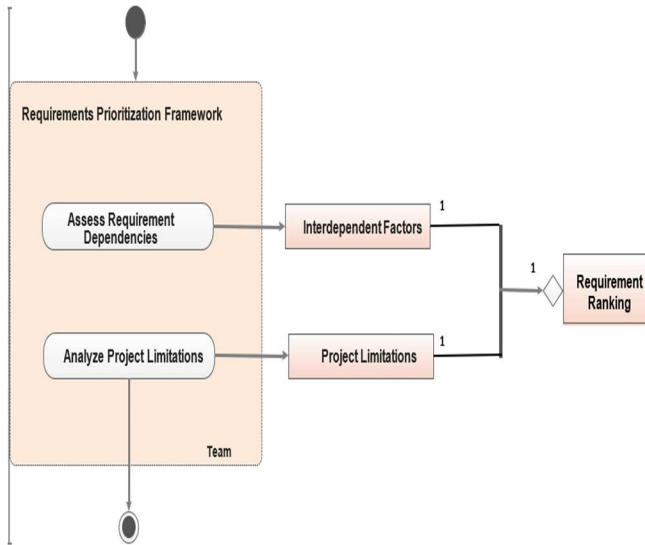


Fig 5. Framework for requirements prioritization

V. CONCLUSION

This study advances the understanding of effective user requirement elicitation in software development by identifying key factors that enhance the Requirements Engineering (RE) process in both Agile and traditional methodologies. Through a survey-based empirical investigation of 50 software companies, the research establishes that Client Engagement (V8), Requirement Prioritization (V9), and Project Planning & Strategy (V12) are statistically significant predictors of project success, as evidenced by correlation, regression, and ANOVA analyses. These findings underpin the proposed Requirement Definition and Evaluation Model, which provides a structured, adaptable framework for managing evolving user needs across development methodologies.

A. Contributions to Theory

The study contributes to RE literature by empirically validating the critical role of client engagement, requirement prioritization, and strategic planning in achieving successful project outcomes. It extends prior work [5,8] by quantifying the impact of these factors through statistical analysis, offering a robust evidence base for their significance. The proposed model integrates insights from both Agile and traditional RE, addressing gaps in existing frameworks by providing a multi-level abstraction at Levels A, B, and C that balances flexibility and structure. This model advances theoretical understanding by formalizing a process for requirement elicitation and analysis that accommodates dynamic user needs, particularly in addressing challenges like miscommunication and scope creep identified in prior studies [3,12].

B. Contributions to Practice

Practically, the Requirement Definition and Evaluation Model offers software development teams a standardized,

methodology-agnostic approach to improve RE processes. By emphasizing client engagement, the model encourages active stakeholder collaboration, reducing the risk of misunderstood requirements and enhancing user satisfaction [8, 9]. The focus on iterative requirement prioritization and strategic planning enables practitioners to adapt to changing requirements efficiently, minimizing delays and cost overruns. The model's use of UML activity diagrams and tools like the Prioritized Requirement Backlog and Sprint Development Backlog provides actionable guidance for project managers to streamline elicitation, validation, and implementation. This is particularly valuable for distributed teams facing communication gaps or limited client availability [12].

C. Limitations

Despite its contributions, the study has limitations. The sample size of 50 companies, while yielding a 66.6% response rate, may limit generalizability across diverse industries or larger populations. The reliance on survey data introduces potential response bias, as participants may not fully represent the perspectives of all stakeholders. Additionally, the study focused on a subset of parameters, potentially overlooking other factors such as non-functional requirements (NFRs) or cultural influences on RE effectiveness.

D. Future Research Directions

Future research should address these limitations by expanding the sample size and diversity to include various industries and project scales. Investigating the role of NFRs in Agile and traditional RE could enhance the model's robustness, building on solutions like agile use cases. Exploring the impact of cultural and organizational factors on client engagement and requirement prioritization may provide deeper insights into distributed team dynamics. Additionally, longitudinal studies could validate the model's effectiveness over multiple project cycles, assessing its adaptability to emerging technologies and development paradigms.

By addressing these theoretical and practical dimensions, this study provides a foundation for improving RE processes and encourages further exploration to enhance software development outcomes.

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